

16.8.1 Problems

16.1 In Examples 16.2 and 16.4, we presented the linear probability and probit model estimates using an example of transportation choice. The logit model for the same example is $P(AUTO = 1) = \Lambda(\gamma_1 + \gamma_2 DTIME)$, where $\Lambda(\bullet)$ is the logistic *cdf* in equation (16.7). The logit model parameter estimates and their standard errors are

$$\begin{array}{ccc} \tilde{\gamma}_1 + \tilde{\gamma}_2 DTIME = -0.2376 + 0.5311 DTIME \\ (se) & (0.7505) & (0.2064) \end{array}$$

- Calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$.
- Using the probit model results in Example 16.4, calculate the estimated probability that a person will choose automobile transportation given that $DTIME = 1$. How does this result compare to the logit estimate? [Hint: Recall that Statistical Table 1 gives cumulative probabilities for the standard normal distribution.]
- Using the logit model results, compute the estimated marginal effect of an increase in travel time of 10 minutes for an individual whose travel time is currently 30 minutes longer by bus (public transportation). Using the linear probability model results, compute the same marginal effect estimate. How do they compare?
- Using the logit model results, compute the estimated marginal effect of a decrease in travel time of 10 minutes for an individual whose travel time is currently 50 minutes longer by driving. Using the probit results, compute the same marginal effect estimate. How do they compare?

(a)

$$z = -0.2376 + 0.5311 * 1 = 0.2935$$

$$P(AUTO = 1 / DTIME = 1) = 1 / \{1 + e^{-0.2935}\} = 57.3\%$$

(b)

$$z = -0.0644 + 0.3 * 1 = 0.2356$$

$$\Phi(0.2356) = 59.31\%$$

(c)

$$z_1 = -0.2376 + 0.5311 * 30 = 15.6954$$

$$P_1 = 1 / \{1 + e^{-15.6954}\} = 1$$

$$z_2 = -0.2376 + 0.5311 * 40 = 21.0064 \text{ implies}$$

$$P_2 = 1 / \{1 + e^{-21.0064}\} = 1.0000$$

LPM (線性機率模型) 的邊際效應

$$\Delta P = \beta_{LPM} * \Delta DTIME = 0.0703 * 10 = 0.703$$

兩者結果存在實質且巨大的差異，這正好凸顯了線性機率模型 (LPM) 在極端值下的致命缺陷。Logit 模型呈現了合理的『邊際效益遞減』特性

(d)

Logit 模型的計算結果

當 DTIME = -50 時：

$$z_1 = -0.2376 + 0.5311 * (-50) = -26.79$$

$$P_1 = 1 / \{1 + e^{\{26.79\}}\} = 0$$

DTIME = -40 時：

$$z_2 = -0.2376 + 0.5311 * (-40) = -21.48$$

$$P_2 = 1 / \{1 + e^{\{21.48\}}\} = 0$$

probit 模型的計算結果

當 DTIME = -50 時：

$$z_1 = -0.0644 + 0.3 * (-50) = -15.06$$

$$P_1 = \Phi(-15.06) = 0$$

當 DTIME = -40 時：

$$z_2 = -0.0644 + 0.3 * (-40) = -12.06$$

$$P_2 = \Phi(-12.06) = 0$$

Logit 與 Probit 模型算出的結果高度一致，兩者得到的邊際效應皆趨近於 0。

這個結果非常符合現實直覺。因為一開始開車時間比公車慢了足足 50 分鐘（DTIME = -50），這是一個極端惡劣的狀況，此時選擇開車的機率本來就已經飽和收斂在 0% 左右。

即便開車時間縮短了 10 分鐘（變成慢 40 分鐘），開車依然是一件極度沒有效率的事，完全無法動搖人類的決策。

